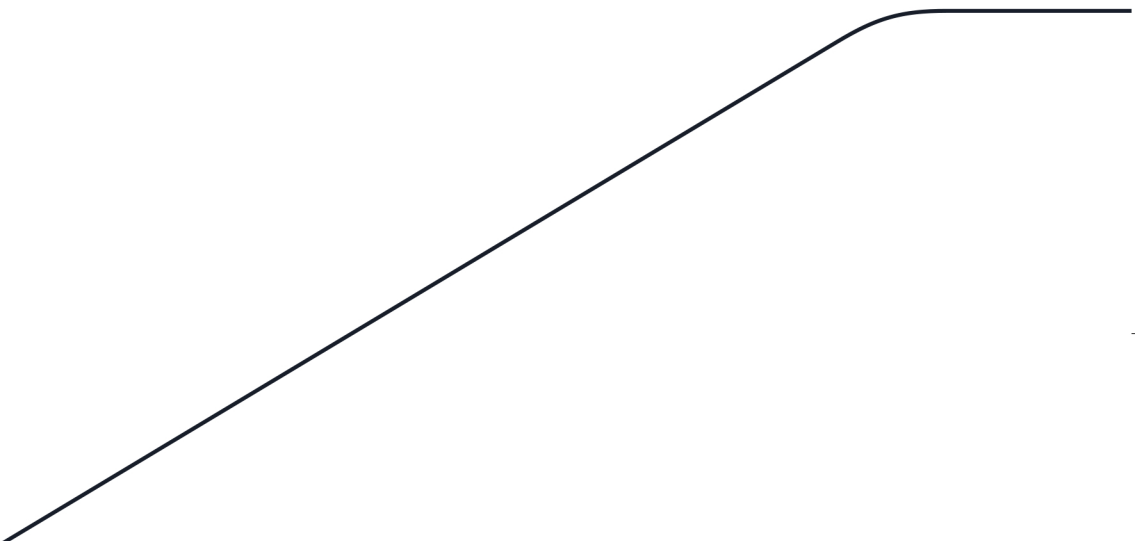


2024

Forte

Owner's Manual



Movement that inspires

Disc brakes wear indicator

Your vehicle has disc brakes.

When your brake pads are worn and new pads are required, you will hear a high-pitched warning sound from your front or rear brakes. You may hear this sound come and go or it may occur whenever you depress the brake pedal.

⚠ CAUTION

- To avoid costly brake repairs, do not continue to drive with worn brake pads.
- Always replace the front or rear brake pads as pairs.

⚠ WARNING

Brake wear

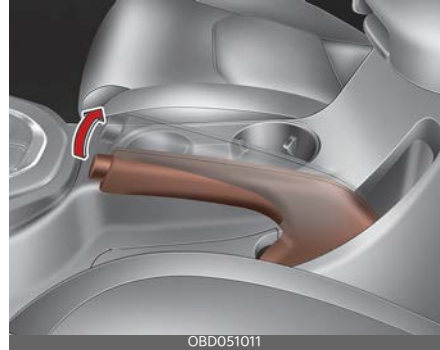
The brake wear warning sound means your vehicle needs service. If you ignore this audible warning, you will eventually lose braking performance, which could lead to a serious accident.

Rear drum brakes (if equipped)

Your rear drum brakes do not have wear indicators. Therefore, have the rear brake linings inspected if you hear a rear brake rubbing noise. Have your rear brakes inspected each time you change or rotate your tires and when you have the front brakes replaced.

Parking brake (Hand-type) (if equipped)

Applying the parking brake



To engage the parking brake, first apply the foot brake and then pull up the parking brake lever as far as possible.

When parking on an incline, the shift lever should be in a low gear (manual transmission) vehicles.

⚠ CAUTION

- Driving with the parking brake applied will cause excessive brake pad and brake rotor wear.
- Do not operate the parking brake while the vehicle is moving except in an emergency situation. It could damage the vehicle system and endanger driving safety.

Engine compartment.....	8-4
Maintenance services.....	8-5
Owner maintenance	8-6
Scheduled maintenance service	8-8
Explanation of scheduled maintenance items.....	8-15
Checking fluid levels	8-18
Engine oil and filter	8-18
• Checking the engine oil level.....	8-18
• Changing the engine oil and filter	8-19
Engine coolant	8-20
• Checking the coolant level.....	8-20
• Changing the coolant	8-22
Brake/clutch fluid	8-23
• Checking the brake/clutch fluid level	8-23
Washer fluid	8-24
• Checking the washer fluid level.....	8-24
Parking brake (Hand type).....	8-25
• Checking the parking brake.....	8-25
Air cleaner	8-26
• Filter replacement.....	8-26
Climate control air filter	8-27
• Filter inspection.....	8-27
Wiper blades	8-28
• Blade inspection	8-28
• Blade replacement	8-29
Battery	8-31
Tires and wheels	8-33
• Checking tire inflation pressure.....	8-34

8 Maintenance

• Tire rotation	8-35
• Wheel alignment and tire balance	8-36
• Tire replacement	8-36
• Wheel replacement	8-37
• Tire traction	8-37
• Tire maintenance	8-38
• Tire sidewall labeling	8-38
• Tire terminology and definitions	8-41
• All season tires	8-44
• Summer tires	8-44
• Snow tires	8-44
• Tire chains	8-45
• Radial-ply tires	8-45
• Low aspect ratio tire	8-46
Fuses	8-47
• Replacing inner panel fuse	8-48
• Replacing engine compartment fuse	8-49
• Fuse/relay panel description	8-51
Light bulbs	8-58
• Bulb replacement precaution	8-58
• Light bulb position (Front)	8-59
• Light bulb position (Rear)	8-60
• Light bulb position (Side)	8-61
• Headlamp (High/Low) (Bulb type) bulb replacement	8-61
• Front turn signal lamp/Position lamp (Bulb type) bulb replacement	8-62
• Side marker (Bulb type) bulb replacement	8-63
• Position lamp/Daytime running lamp (LED type) bulb replacement	8-63
• Headlamp (LED type) bulb replacement	8-64
• Front fog lamp (LED type) bulb replacement	8-64
• Side repeater lamp (Bulb type) bulb Replacement	8-64

Maintenance

Engine compartment

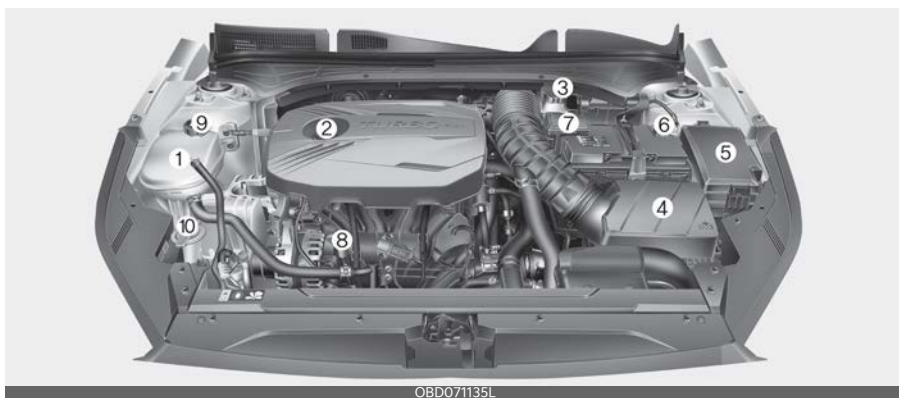
Open the hood to see the engine compartment.

(Gasoline) 2.0 MPI



OBD078101L

(Gasoline) 1.6 T-GDi



OBD071135L

* The actual engine compartment in your Kia may differ from the illustration.

- | | |
|---------------------------------|---------------------------------------|
| 1. Engine coolant reservoir | 8. Engine oil dipstick |
| 2. Engine oil filler cap | 9. Radiator cap |
| 3. Brake/clutch fluid reservoir | 10. Windshield washer fluid reservoir |
| 4. Air cleaner | |
| 5. Fuse box | |
| 6. Negative battery terminal | |
| 7. Positive battery terminal | |

Owner maintenance precautions

Improper or incomplete service may result in problems. This section gives instructions only for the maintenance items that are easy to perform.

As explained earlier in this section, several procedures can be done only by an authorized Kia dealer with special tools.

* NOTICE

Improper owner maintenance during the warranty period may affect warranty coverage. For details, read the separate Warranty & Consumer Information manual provided with the vehicle. If you're unsure about any servicing or maintenance procedure, have it done by an authorized Kia dealer.

⚠ WARNING

Maintenance work

Do not wear jewelry or loose clothing while working under the hood of your vehicle with the engine running. These items can become entangled in moving parts, if you must run the engine while working under the hood, make certain that you remove all jewelry (especially rings, bracelets, watches, and necklaces) and all neckties, scarves, and similar loose clothing before getting near cooling fans.

⚠ WARNING

Touching metal parts

Do not touch metal parts (including strut bars) while the vehicle is operating or hot. Doing so could result in serious bodily injury. Turn the vehicle off and wait until the metal parts cool down to perform maintenance work on the vehicle.

Owner maintenance

The following lists detail the vehicle checks and inspections that should be performed by the owner or an authorized Kia dealer. They should be performed at the indicated frequencies to help ensure the safe and dependable operation of your vehicle.

Any adverse conditions should be brought to the attention of your dealer as soon as possible.

These owner maintenance checks are generally not covered by warranties and you may be charged for labor, parts and lubricants used.

When you stop for fuel:

- Check the coolant level in coolant reservoir.
- Check the windshield washer fluid level.
- Look for low or under-inflated tires. Check if the front of the radiator and condenser are clean and not blocked with leaves, dirt or insects etc. If any of the above parts are extremely dirty or you are not sure of their condition, take your vehicle to an authorized Kia dealer.

⚠ WARNING

Hot coolant

Be careful when checking your engine coolant level when the engine is hot. Scalding hot coolant and steam may blow out under pressure.

While operating your vehicle:

- Check for vibrations in the steering wheel. Notice any increased steering effort or looseness in the steering wheel, or change in its straight-ahead position.
- Notice if your vehicle constantly turns slightly or "pulls" to one side when traveling on smooth, level road.
- When stopping, listen and check for unusual sounds, pulling to one side, increased brake pedal travel or "hard-to-push" brake pedal.
- If any slipping or changes in the operation of your transmission occurs, take your vehicle to an authorized Kia dealer.
- Check the transmission P (Park) function.
- Check the parking brake.
- Check for fluid leaks under your vehicle (water dripping from the air conditioning system during or after use is normal).

At least monthly:

- Check the coolant level in the coolant reservoir.
- Check the operation of all exterior lights, including the stoplights, turn signals and hazard warning flashers.
- Check the inflation pressures of all tires including the spare (if provided) for tires that are worn, show uneven wear, or are damaged.
- Check for loose wheel lug nuts.

At least twice a year (i.e., every Spring and Fall):

- Check the radiator, heater and air conditioning hoses for leaks or damage.
- Check the windshield washer spray and wiper operation. Clean the wiper blades with clean cloth dampened with washer fluid.
- Check the headlight alignment.
- Check the lap/shoulder belts for wear and function.

At least once a year:

- Clean the body and door drain holes.
- Lubricate the door hinges and check the hood hinges.
- Lubricate the door and hood locks and latches.
- Lubricate the door rubber weatherstrips.
- Check the air conditioning system.
- Inspect and lubricate transmission linkage and controls.
- Clean the battery and terminals.
- Check the brake fluid level.
- Visually inspect steering, suspension, and chassis components for damaged, loose, or missing parts or signs of wear.

Normal maintenance schedule - Non Turbo Models

The following maintenance services must be performed to ensure good emission control and performance. Keep receipts for all vehicle services to protect your warranty. Where both mileage and date are shown, the frequency of service is determined by whichever occurs first.

I: Inspect and if necessary, adjust, correct, clean or replace.

R: Replace or change

Number of months or driving distance, whichever comes first													
Months		12	24	36	48	60	72	84	96	108	120	132	144
Miles×1,000		8	16	24	32	40	48	56	64	72	80	88	96
Km×1,000		13	26	39	52	65	78	91	104	117	130	143	156
Tire rotation		Rotate every 8,000 miles (13,000 km) or 12 months											
Engine oil and engine oil filter ¹	(Gasoline) 2.0 MPI	R	R	R	R	R	R	R	R	R	R	R	R
Climate control air filter		I	R	I	R	I	R	I	R	I	R	I	R
Air cleaner filter (Engine)		I	I	R	I	I	R	I	I	R	I	I	R
Brake/clutch fluid		Inspect every 8,000 miles (13,000 km) or 12 months Replace every 48,000 miles (78,000 km) or 48 months											
Spark plugs	(Gasoline) 2.0 MPI	Replace every 96,000 miles (156,000 km)											
Coolant (Engine) ²		At first, replace at 120,000 miles (195,000 km) or 120 months After that, replace every 24,000 miles (39,000 km) or 24 months											
12V Battery condition		I	I	I	I	I	I	I	I	I	I	I	I
Vacuum hose													
Air conditioner refrigerant													
Exhaust system													
Brake lines, hoses and connections													
Brake discs and pads													
Suspension ball joints and mounting bolts													
Steering gear rack, linkage and boots													
Air conditioner compressor													
Parking brake (Hand type)		-	I	-	I	-	I	-	I	-	I	I	
Drive shaft and boots													
Fuel tank and fuel cap													
Fuel lines, hoses and connections													
Fuel tank air filter													
Cooling system		-	-	-	I	-	I	-	I	-	I	-	I
Drive belts ³		-	-	-	-	-	-	I	I	I	I	I	I
Manual transmission (MT) fluid ⁴		No service required											
Intelligent Variable Transmission (IVT) fluid ⁵													

*1. As it is normal for engine oil to be consumed during driving, the engine oil level should be checked on regular basis.

The engine oil change interval for normal operating conditions is based on the use of the recommended engine specification. If the recommended engine oil specification is not used, then replace the engine oil according to the maintenance schedule under severe operating conditions.

Never add any additives to the engine oil. Engine oil additives can change the properties of engine oil and may cause serious engine failure.

*2. When adding coolant, use only deionized water or soft water for your vehicle and never mix hard water in the coolant filled at the factory. An improper coolant mixture can result in serious malfunction or engine damage.

*3. Inspect alternator, water pump and air conditioner drive belt and if necessary, repair or replace. Inspect drive belt tensioner, idler and alternator pulley and if necessary correct or replace.

* If the vehicle has been submerged in water or in a flooded area, the fluids should be changed as a precaution.

- Fuel filter (gasoline engine): The fuel filter is considered to be maintenance free but periodic inspection is recommended for this maintenance schedule depends on fuel quality.
 - If there are some important problems like fuel flow restriction, surging, loss of power, hard starting problem etc., replace the fuel filter immediately regardless of maintenance schedule and consult an authorized Kia dealer for details.

Maintenance Under Severe Usage Conditions - Non Turbo Models

The following items must be serviced more frequently on cars mainly used under severe driving conditions. Refer to the chart below for the appropriate maintenance intervals.

I: Inspect and if necessary, adjust, correct, clean or replace
R: Replace

Maintenance item		Maintenance operation	Maintenance intervals	Driving condition
Engine oil and engine oil filter	(Gasoline) 2.0 MPI	R	Every 5,000 miles (8,000 km) or 6 months	A, B, C, D, E, F, G, H, I, J, K
Intelligent Variable Transmission (IVT) fluid		R	Every 56,000 miles (91,000 km)	A, C, F, G, H, I, J, K
Manual transmission (MT) fluid		R	Every 72,000 miles (117,000 km)	C, D, F, G, H, I, J
Air cleaner filter		R	More frequently	C, E
Spark plugs		R	More frequently	A, B, F, G, H, I, K
Climate control air filter		I	More frequently	C, E, G
Parking brake (Hand type)		I	More frequently	C, D, G, H
Brake discs, pads and calipers		I	More frequently	C, D, E, G, H, I, J, K
Suspension ball joints and mounting bolts		I	More frequently	C, D, E, G, H, I
Steering gear rack, linkage and boots		I	More frequently	C, D, E, F, G
Drive shafts and boots		I	More frequently	C, D, E, F, G, H, I, J

Severe Driving Conditions

A: Repeatedly driving short distance of less than 5 miles (8 km) in normal temperature or less than 10 miles (16 km) in freezing temperature.

B: Extensive low speed driving for long distances.

C: Driving on rough, dusty, muddy, unpaved, graveled or salt-spread roads.

D: Driving in areas using salt or other corrosive materials or in very cold weather.

E: Driving in heavy dust condition.

F: Driving in heavy traffic area.

G: Driving on uphill, downhill, or mountain roads.

H: Using for towing or camping and driving with loading on the roof

I: Driving as a patrol car, taxi, other commercial use or vehicle towing.

J: Frequently driving under high speed or rapid acceleration/deceleration.

K: Frequently driving in stop-and-go conditions

Explanation of scheduled maintenance items

The following parts require scheduled maintenance.

Engine oil and filter

The engine oil and filter should be changed at the intervals specified in the maintenance schedule. If the vehicle is being driven in severe conditions, more frequent oil and filter changes are required.

Drive belts

Inspect all drive belts for evidence of cuts, cracks, excessive wear or oil saturation and replace if necessary. Drive belts should be checked periodically for proper tension and adjusted as necessary.

Fuel filter

Kia gasoline vehicles are equipped with a lifetime fuel filter that is integrated with the fuel tank. Regular maintenance or replacement are generally not needed. This may vary depending on fuel quality. If you experience any of the following: fuel flow restriction, surging, loss of power, or a hard starting issue, inspection and, if necessary, replacement may be needed. Have the fuel filter inspected or replaced by an authorized Kia dealer.

Fuel lines, fuel hoses and connections

Check the fuel lines, fuel hoses and connections for leakage and damage. Have an authorized Kia dealer replace any damaged or leaking parts immediately.

Fuel tank and fuel cap

The fuel tank and fuel cap should be inspected at those intervals specified in the maintenance schedule. Make sure that a new fuel tank or fuel cap is correctly replaced.

Vacuum crankcase ventilation hoses

Inspect the surface of hoses for evidence of heat and/or mechanical damage. Hard and brittle rubber, cracking, tears, cuts, abrasions, and excessive swelling indicate deterioration. Particular attention should be paid to examine those hose surfaces nearest to high heat sources, such as the exhaust manifold. Inspect the hose routing to ensure that the hoses do not come in contact with any heat source, sharp edges or moving components which might cause heat damage or mechanical wear. Inspect all hose connections, such as clamps and couplings, to make sure they are secure, and that no leaks are present. Hoses should be replaced immediately if there is any evidence of deterioration or damage.

Air cleaner filter

A Genuine Kia air cleaner filter is recommended when the filter is replaced.

Spark plugs

Make sure to install new spark plugs of the correct heat range.

When assembling parts, be sure to wipe the inside and outside of the boot bottom of the ignition coil and the insulator of the spark plug with a soft cloth to prevent contamination of the spark plug insulator.

Valve clearance (if equipped)

Inspect for excessive valve noise and/or engine vibration and adjust if necessary. An authorized Kia dealer should perform the operation.

Cooling system

Check the cooling system components, such as the radiator, coolant reservoir, hoses and connections for leakage and damage. Replace any damaged parts.

Coolant

The coolant should be changed at the intervals specified in the maintenance schedule.

Dual clutch transmission (DCT) fluid

Inspect the dual clutch transmission fluid according to the maintenance schedule.

Intelligent Variable Transmission (IVT) fluid

The Intelligent Variable Transmission (IVT) fluid should not be checked under normal usage conditions. But in severe conditions, the fluid should be changed at an authorized Kia dealer in accordance to the scheduled maintenance at the beginning of this chapter. (Refer to the "Maintenance Under Severe Usage Conditions - Non Turbo Models" on page 8-14.)

* NOTICE

The Intelligent Variable Transmission (IVT) fluid color is amber. As the vehicle is driven, the Intelligent Variable Transmission (IVT) fluid will begin to look darker. This is normal, and you should

not judge the need to replace the fluid based upon the changed color.

⚠ CAUTION

If the genuine oil that is developed for best performance is not used, it may cause the problems of transmission performance. (Refer to the "Recommended lubricants and capacities" on page 9-7.)

WARNING

- Do not change oil.
- In case of repair, use only Genuine SP-CVT1
- Ne changez pas l'huile.
- En cas de réparation, utilisez uniquement SP-CVT1
- 请勿随便更换油。
- 需要维修的情况下，只能使用正品 SP-CVT1
- Масло не заменять. В случае ремонта следует использовать только оригинальное SP-CVT1

OBDO78066

Manual transmission (MT) fluid

Inspect the manual transmission fluid according to the maintenance schedule.

Brake hoses and lines

Visually check for proper installation, chafing, cracks, deterioration and any leakage. Replace any deteriorated or damaged parts immediately.

* NOTICE

NHTSA Safety Corrosion Alert

NHTSA has warned all vehicle owners of all brands that they must maintain their vehicles in a manner which will prevent brake hose and brake line failures due to corrosion when such vehicles are exposed to winter road salt and related chemicals. While serious corrosion con-

ditions typically only manifest themselves as safety issues after 7 years of vehicle use, the corrosion process starts immediately and thus underbody cleaning maintenance must commence from your vehicle's first exposure to road salts and chemicals. NHTSA urges vehicle owners to take the following steps to prevent corrosion:

1. Wash the undercarriage of your vehicle regularly throughout the winter and do a thorough washing in the spring to remove road salt and other de-icing chemicals.
 2. Monitor the brake system for signs of corrosion by having regular professional inspections and watching for signs of problems, including loss of brake fluid, unusual leaks and soft or spongy feel in the brake pedal.
 3. Replace the entire brake pipe assembly if you find severe corrosion that causes scaling or flaking of brake components.
-

Brake/clutch fluid

Check the brake/clutch fluid level in the brake/clutch fluid reservoir. The level should be between "MIN" and "MAX" marks on the side of the reservoir. Use only hydraulic brake/clutch fluid conforming to DOT 4 specification.

Parking brake

Inspect the parking brake system including the parking brake lever (or pedal) and cables.

Exhaust pipe and muffler

Visually inspect the exhaust pipes, muffler and hangers for cracks, deterioration, or damage. Start the engine and listen carefully for any exhaust gas leakage. Tighten connections or replace parts as necessary.

Brake discs, pads and calipers

Check the pads for excessive wear, discs for run out and wear, and calipers for fluid leakage.

Suspension mounting bolts

Check the suspension connections for looseness or damage. Retighten to the specified torque.

Steering gear box, linkage & boots/lower arm ball joint

With the vehicle stopped and off, check for excessive free-play in the steering wheel.

Check the linkage for bends or damage. Check the dust boots and ball joints for deterioration, cracks, or damage. Replace any damaged parts.

Drive shafts and boots

Check the drive shafts, boots and clamps for cracks, deterioration, or damage. Replace any damaged parts and, if necessary, repack the grease.

Air conditioning refrigerant

Check the air conditioning lines and connections for leakage and damage.

Checking fluid levels

When checking engine oil, engine coolant, brake fluid, and washer fluid, always be sure to clean the area around any filler plug, drain plug, or dipstick before checking or draining any lubricant or fluid. This is especially important in dusty or sandy areas and when the vehicle is used on unpaved roads. Cleaning the plug and dipstick areas will prevent dirt and grit from entering the engine and other mechanisms that could be damaged.

Engine oil and filter

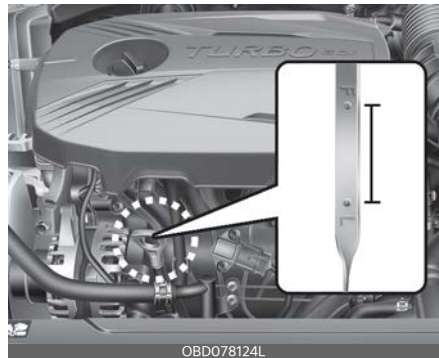
Checking the engine oil level

Engine oil is used for lubricating, cooling, and operating various hydraulic components in the engine. Engine oil consumption while driving is normal, and it is necessary to check and refill the engine oil regularly. Also, check and refill the oil level within the recommended maintenance schedule to prevent deterioration of oil performance. Check the engine oil following the below procedure.

(Gasoline) 2.0 MPI



(Gasoline) 1.6 T-GDi



1. Be sure the vehicle is on level ground.
2. Start the engine and allow it to reach normal operating temperature.

3. Turn the engine off, remove the oil filler cap and pull the dipstick out. Wait for 15 minutes for the oil to return to the oil pan.
4. Wipe the dipstick clean and re-insert it fully.
5. Pull the dipstick out again and check the level. Check if the oil level is between the F-L line, and if it is below the L line, add enough oil to bring the level to F line.

WARNING

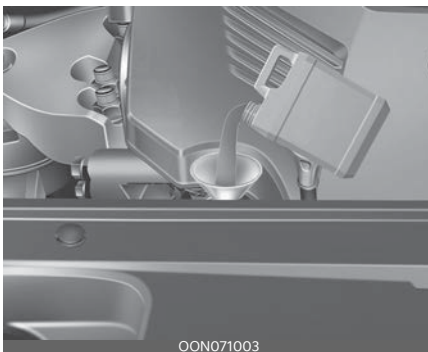
Radiator hose

Be very careful not to touch the radiator hose when checking or adding engine oil as it may be hot enough to burn you.

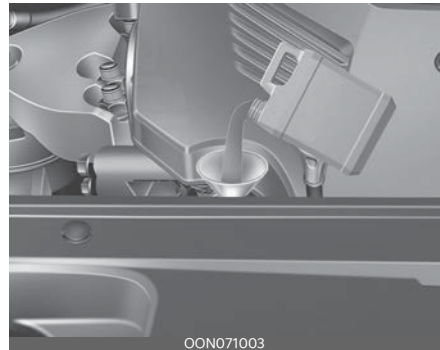
CAUTION

When you wipe the oil level gauge, you should wipe it with a clean cloth. When mixed with debris, it can cause engine damage.

(Gasoline) 2.0 MPI



(Gasoline) 1.6 T-GDi



Use a funnel to help prevent oil from being spilled on engine components.

Use only the specified engine oil.

(Refer to "Recommended lubricants and capacities" on page 9-7.)

- Do not spill engine oil when adding or changing engine oil. Wipe off spilled oil immediately.
- The engine oil consumption may increase while you break in a new vehicle and it will be stabilized after driving 4,000 miles (6,000 km).
- The engine oil consumption can be affected by driving habits, climate conditions, traffic conditions, oil quality, etc. Therefore, it is recommended that you inspect the engine oil level regularly and refill it if necessary.

Changing the engine oil and filter

Have engine oil and filter changed by an authorized Kia dealer according to the Maintenance Schedule at the beginning of this chapter.

- If the maintenance schedule to replace engine oil is exceeded, the engine oil performance may deteriorate, and the engine condition may be affected. Therefore, replace the engine oil according to the maintenance schedule.

- To keep the engine in optimal condition, use the recommended engine oil and filter. If the recommended engine oil and filter are not used replace it according to the maintenance schedule under severe usage conditions.
- The purpose of the maintenance schedule for engine oil replacement is to prevent oil deterioration and it is irrelevant to oil consumption. Check and refill engine oil regularly.

⚠ WARNING

Used engine oil may cause irritation or cancer of the skin if left in contact with the skin for prolonged periods of time. Used engine oil contains chemicals that have caused cancer in laboratory animals. Always protect your skin by washing your hands thoroughly with soap and warm water as soon as possible after handling used oil.

Do not leave used engine oil within the reach of children.

⚠ CAUTION

The engine oil is very hot immediately after the vehicle has been driven and can cause burns during replacement. Replace the engine oil after the engine oil has cooled down.

* NOTICE

Never add any additives to the engine oil. Engine oil additives can change the properties of engine oil and may cause serious engine failure.

Engine coolant

The high-pressure cooling system has a reservoir filled with year round anti-freeze coolant. The reservoir is filled at the factory.

Check the antifreeze protection and coolant level at least once a year, at the beginning of the winter season, and before traveling to a colder climate.

Checking the coolant level

⚠ WARNING



Radiator cap

Never attempt to remove the radiator or inverter cap while the engine is operating or

hot. Doing so might lead to cooling system and engine damage and could result in serious bodily injury from escaping hot coolant or steam.

- Turn the engine off and wait until it cools down. Use extreme care when removing the radiator cap. Wrap a thick towel around it, and turn it counterclockwise slowly to the first stop. Step back while the pressure is released from the cooling system. When you are sure all the pressure has been released, press down on the cap, using a thick towel, and continue turning counterclockwise to remove it.
- Even if the engine is not operating, do not remove the radiator cap or the drain plug while the engine and radiator are hot. Hot coolant and steam may still blow out under pressure, causing serious injury.

⚠ WARNING

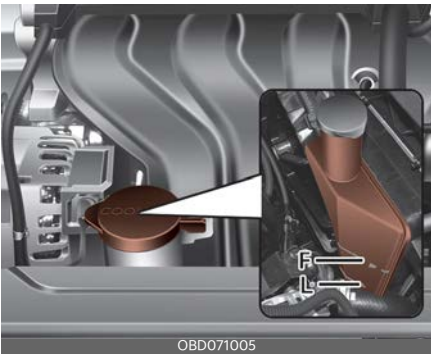


Cooling fan

Use caution when working near the blade of the cooling fan. The electric motor

(cooling fan) is controlled by engine coolant temperature, refrigerant pressure and vehicle speed. It can operate even when the engine is not running.

(Gasoline) 2.0 MPI



(Gasoline) 1.6 T-GDi



Check the condition and connections of all cooling system hoses and heater hoses. Replace any swollen or deteriorated hoses.

The coolant level should be filled between F and L (or MAX and MIN) marks on the coolant level gauge when the engine is cool.

If the coolant level is low, add enough specified coolant to provide protection against freezing and corrosion. Bring the level to F (or MAX), but do not overfill. If frequent additions are required, see an authorized Kia dealer for a cooling system inspection.

*** NOTICE**

Make sure the coolant cap is properly closed after refill of coolant. Otherwise the engine could be overheated while driving.

1. Check if the radiator cap label is on straight.

Engine room front view



2. Make sure that the tiny protrusions inside the coolant cap are securely interlocked.

Engine room rear view



Recommended engine coolant

- When adding coolant, use only deionized water or soft water for your vehicle and never mix hard water in the coolant filled at the factory. An improper coolant mixture can result in serious malfunction or engine damage.
- The engine in your vehicle has aluminum engine parts and must be protected by an ethylene-glycol with phosphate based coolant to prevent corrosion and freezing.
- DO NOT USE alcohol or methanol coolant or mix them with the specified coolant.
- Do not use a solution that contains more than 60% antifreeze or less than 35% antifreeze. This would reduce the effectiveness of the solution.

For mixture percentage, refer to the following table.

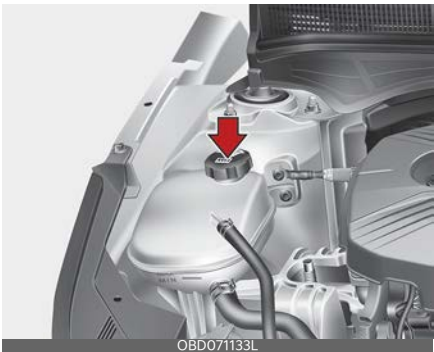
Ambient Temperature	Mixture Percentage (volume)	
	Antifreeze	Water
5°F (-15°C)	35	65
-13°F (-25°C)	40	60
-31°F (-35°C)	50	50

Ambient Temperature	Mixture Percentage (volume)	
	Antifreeze	Water
-49°F (-45°C)	60	40

(Gasoline) 2.0 MPI



(Gasoline) 1.6 T-GDi



WARNING



Radiator cap

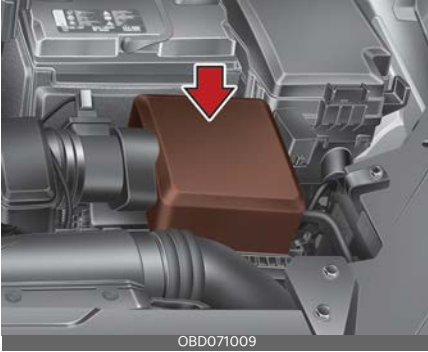
Do not remove the radiator cap when the engine and radiator are hot. Scalding hot coolant and steam may blow out under pressure which may result in serious injury.

Changing the coolant

Have the coolant changed by an authorized Kia dealer according to the Maintenance

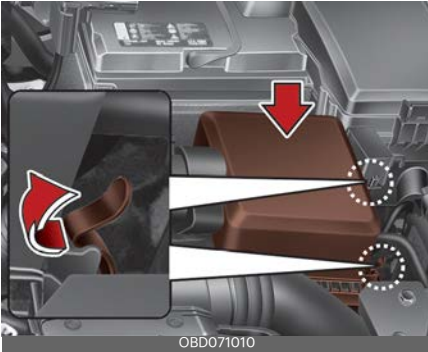
Air cleaner

Filter replacement

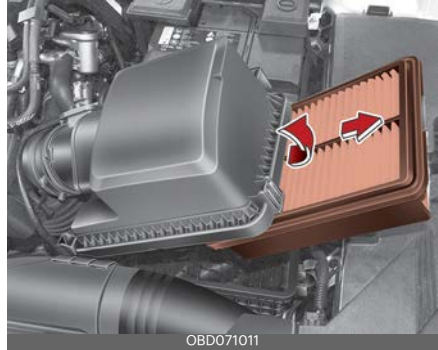


It must be replaced when necessary, and should not be washed.

1. Loosen the air cleaner cover attaching clips and open the cover.



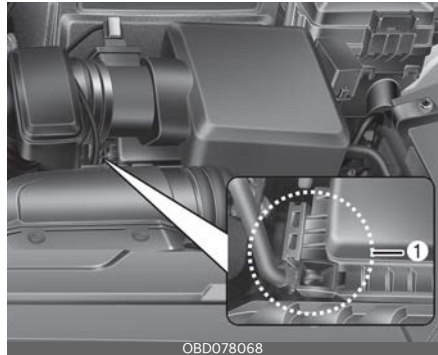
2. Wipe the inside of the air cleaner.



3. Replace the air cleaner filter.
4. Lock the cover with the cover attaching clips.

* NOTICE

Insert the hinge(1) and engage the clips when mounting the air cleaner cover.



Replace the filter according to the Maintenance Schedule.

If the vehicle is operated in extremely dusty or sandy areas, replace the element more often than the usual recommended intervals. (Refer to "Recommended lubricants and capacities" on page 9-7.)

⚠ CAUTION

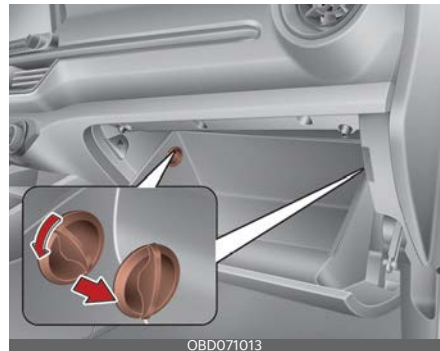
- Do not drive with the air cleaner removed; this will result in excessive engine wear.
- When removing the air cleaner filter, be careful that dust or dirt does not enter the air intake, or damage may result.
- Use a Kia genuine part. Use of a non-genuine part could damage the air flow sensor.

Climate control air filter

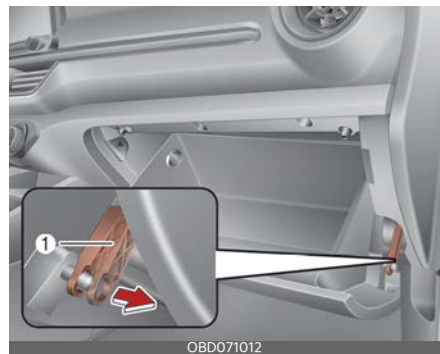
Filter inspection

The climate control air filter should be replaced according to the maintenance schedule. If the vehicle is operated in severely air-polluted cities or on dusty rough roads for a long period, it should be inspected more frequently and replaced earlier. When you replace the climate control air filter, replace it by performing the following procedure, and be careful to avoid damaging other components.

1. Open the glove box and remove the stoppers on both sides.



2. With the glove box open, pull the support strap (1).



Tires and wheels

For proper maintenance, safety, and maximum fuel economy, you must always maintain the recommended tire inflation pressures and stay within the load limits and weight distribution recommended for your vehicle.

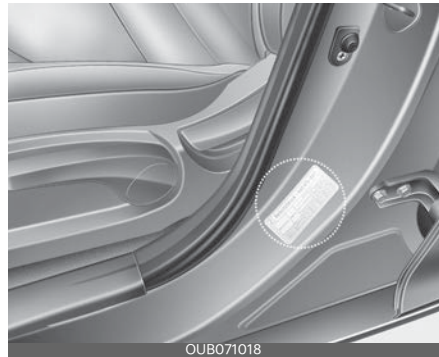
Recommended cold tire inflation pressures

All tire pressures should be checked when the tires are cold. "Cold Tires" means the vehicle has not been driven for at least three hours or driven less than 1 mile (1.6 km).

Recommended pressures must be maintained for the best ride, vehicle handling, and minimum tire wear.

For recommended inflation pressure, refer to "Tires and wheels" on page 9-6.

All specifications (sizes and pressures) can be found on a label attached to the driver's side center pillar.



WARNING

Tire underinflation

Inflate your tires consistent with the instructions provided in this manual. Regularly check the tire inflation pressure, and correct it as needed. Driving on under-inflated tires not only compromises your vehicle's driving stability but also may lead to tire damage and the risk of an accident. Severe under inflation (10 psi (70 kPa) or more) can lead to severe heat build up, causing blowouts, tread separation and other tire failures that can result in the loss of vehicle control. This risk is much higher on hot days and when driving for long periods at high speeds.

Failure to maintain specified pressure may result in excessive wear, poor handling, reduced fuel economy, deformation of tire and/or wheel, harsh ride con-

ditions, possibility for additional damage from road hazards, or result in tire failure.

Tire pressure

Always observe the following:

- Check tire pressure when the tires are cold. (After vehicle has been parked for at least three hours or hasn't been driven more than 1 mile (1.6 km) since startup.)
- Check the pressure of your spare tire each time you check the pressure of other tires.
- Never overload your vehicle. Be careful not to overload a vehicle luggage rack if your vehicle is equipped with one.
- Warm tires normally exceed recommended cold tire pressures by 4~6 psi (28~41 kPa). Do not release air from warm tires to adjust the pressure or the tires will be underinflated.

WARNING

Tire Inflation

Overinflation or underinflation can reduce tire life, adversely affect vehicle handling, and lead to sudden tire failure. This could result in loss of vehicle control and potential injury.

Checking tire inflation pressure

Check your tires once a month or more.

Use a good quality gauge to check tire pressure. You cannot tell if your tires are properly inflated simply by looking at them. Radial tires may look properly inflated even when they're underinflated.

Check the tire inflation pressure when the tires are cold. "Cold" means your vehicle has been sitting or at least three hours or driven no more than 1 mile (1.6 km).

1. Remove the valve cap from the tire valve stem.
2. Press the tire gauge firmly onto the valve to get a pressure measurement. If the cold tire inflation pressure matches the recommended pressure on the tire and loading information label, no further adjustment is necessary.
3. If the pressure is low, add air until you reach the recommended amount.
4. If you overfill the tire, release air by pushing on the metal stem in the center of the tire valve.
5. Recheck the tire pressure with the tire gauge.

6. Be sure to put the valve caps back on the valve stems. They help prevent leaks by keeping out dirt and moisture.

Inspect your tires frequently for proper inflation as well as wear and damage. Always use a tire pressure gauge.

Tires with too much or too little pressure wear unevenly. This could result in poor handling, loss of vehicle control, and sudden tire failure leading to accidents, injuries, and even death. The recommended cold tire pressure for your vehicle can be found in this manual and on the tire label located on the driver's side center pillar.

Tire rotation

To equalize tread wear, it is recommended that the tires be rotated every 6,000 miles (10,000 km) (for Non-Turbo engine), 8,000 miles (13,000 km) (for Non Turbo engine) or sooner if irregular wear develops.

During rotation, check the tires for correct balance.

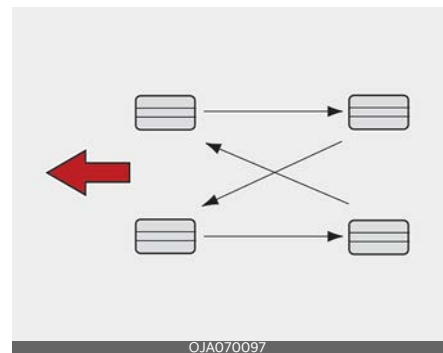
When rotating tires, check for uneven wear and damage. Abnormal wear is usually caused by incorrect tire pressure, improper wheel alignment, out of-balance wheels, severe braking or severe cornering. Look for

bumps or bulges in the tread or side of tire. Replace the tire if you find either of these conditions. Replace the tire if fabric or cord is visible. After rotation, be sure to bring the front and rear tire pressures to specification and check lug nut tightness. (proper torque is 79~94 lb·ft [11~13 kgf·m])

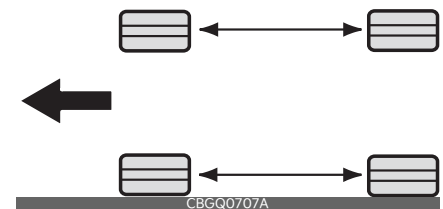
Refer to "Tires and wheels" on page 9-6.

Disc brake pads should be inspected for wear whenever tires are rotated.

Without a spare tire



Directional tires (if equipped)



Rotate radial tires that have an asymmetric tread pattern only from front to rear and not from right to left.

⚠ WARNING

Mixing tires

- Do not use the compact spare tire (if equipped) for tire rotation.
- Do not mix bias ply and radial ply tires under any circumstances. This may cause unusual handling characteristics.

Wheel alignment and tire balance

The wheels on your vehicle were aligned and balanced carefully at the factory to give you the longest tire life and best overall performance.

In most cases, you will not need to have your wheels aligned again. However, if you notice unusual tire wear or your vehicle pulling one way or the other, the alignment may need to be reset. If you notice your vehicle vibrating when driving on a smooth road, your wheels may need to be rebalanced.

⚠ CAUTION

Wheel weight

Improper wheel weights can damage your vehicle's aluminum wheels. Use only approved wheel weights.

Tire replacement

If the tire is worn evenly, a tread wear indicator (A) will appear as a solid band across the tread.



This shows there is less than 1/16 inch (1.6 mm) of tread left on the tire. Replace the tire when this happens.

Do not wait for the band to appear across the entire tread before replacing the tire.

The Anti-lock Brake System (ABS) works by comparing the speed of the wheels. The tire size affects wheel speed. When replacing tires, all 4 tires must use the same size originally supplied with the vehicle. Using tires

of a different size can cause the ABS and Electronic Stability Control (ESC) to work irregularly. It is best to replace all four tires at the same time. If that is not possible, or necessary, then replace the two front or two rear tires as a pair. Replacing just one tire can seriously affect your vehicle's handling.

*** NOTICE**

We recommend that when replacing tires, use the same originally supplied with the vehicles. If not, that affects driving performance.

Compact spare tire replacement (if equipped)

A compact spare tire has a shorter tread life than a regular size tire. Replace it when you can see the tread wear indicator bars on the tire. The replacement compact spare tire should be the same size and design tire as the one provided with your new vehicle and should be mounted on the same compact spare tire wheel. The compact spare tire is not designed to be mounted on a regular size wheel, and the compact spare tire wheel is not designed for mounting a regular size tire.

Wheel replacement

When replacing the metal wheels for any reason, make sure the new wheels are equivalent to the original factory units in diameter, rim width and offset.

A wheel with an incorrect size may adversely affect many things: wheel and bearing life, braking and stopping abilities, handling characteristics, ground clearance, body-to-tire clearance, snow chain clearance, speedometer and odometer calibration, headlight aiming and bumper height.

⚠ CAUTION

Wheels

Wheels that do not meet Kia specifications may fit poorly and result in damage to the vehicle or unusual handling and poor vehicle control.

Tire traction

Tire traction can be reduced if you drive on worn tires, tires that are improperly inflated or on slippery road surfaces.

Tires should be replaced when tread wear indicators appear. Slow down whenever there is rain, snow or ice on the road to reduce the possibility of losing control of the vehicle.

Tire maintenance

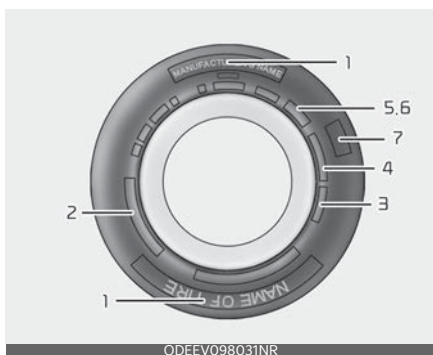
In addition to proper inflation, correct wheel alignment helps to decrease tire wear.

If you find a tire is worn unevenly, have your dealer check the wheel alignment.

When you have new tires installed, make sure they are balanced. This will increase vehicle ride comfort and tire life. Additionally, a tire should always be rebalanced if it is removed from the wheel.

Tire sidewall labeling

This information identifies and describes the fundamental characteristics of the tire and also provides the Tire Identification Number (TIN) for safety standard certification.



The TIN can be used to identify the tire in case of a recall.

1. Manufacturer or brand name

Manufacturer or Brand name is shown.

2. Tire size designation

A tire's sidewall is marked with a tire size designation. You will need this information when selecting replacement tires for your vehicle. The following explains what the letters and numbers in the tire size designation mean.

Example tire size designation:

(These numbers are provided as an example only; your tire size designator could vary depending on your vehicle.)

P235/65R17 108T

- P: Applicable vehicle type (tires marked with the prefix "P" are intended for use on passenger vehicles or light trucks; however, not all tires have this marking).
- 235: Tire width in millimeters.
- 65: Aspect ratio. The tire's section height as a percentage of its width.
- R: Tire construction code (Radial).
- 17: Rim diameter in inches.
- 108: Load Index, a numerical code associated with the maximum load the tire can carry.

Tires and wheels

Item	Tire size	Wheel size	Load Capacity		Speed capacity		Inflation pressure [bar(psi, kPa)]				Wheel lug-nut torque Kgf·m (lbf·ft, N·m)
			LI ^{*1}	Kg	SS ^{*2}	Km/h	Normal load		Maximum load		
							Front	Rear	Front	Rear	
Full size tire	195/65R15	6.0J×15	91	615	H	210	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	11 ~ 13(79 ~ 94,107 ~ 127)
	205/55R16	6.5J×16	91	615	H	210	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	
	225/45R17	7.0J×17	91	615	W	270	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	2.3 (33,230)	
	225/40R18	7.5J×18	88	560	W	270	2.4 (35,240)	2.4 (35,240)	2.4 (35,240)	2.4 (35,240)	
	225/40ZR18	7.5J×18	92	630	Y	300	2.4 (35,240)	2.4 (35,240)	2.4 (35,240)	2.4 (35,240)	
Compact spare tire	T125/80D15	4.0T×15	95	690	M	130	4.2 (60, 420)				
	T125/80D16	4.0T×16	97	730	M	130					

*1 : Load Index
*2 : Speed Symbol

* NOTICE

- It is permissible to add 3psi to the standard tire pressure specification if colder temperatures are expected soon.
Tires typically loose 1psi for every 12°F temperature drop. If extreme temperature variations are expected, recheck your tire pressure as necessary to keep them properly inflated.
- When replacing tires, use the same originally supplied with the vehicles. If not, that affects driving performance.
- When driving in high altitude grades, the atmospheric pressure decreases. Check the tire pressure and add more air when necessary. Additionally required tire air pressure per mile (km) above sea level: 1.5psi/km

⚠ CAUTION

When replacing tires, use the same size originally supplied with the vehicle.
Using tires of a different size can damage related parts or function abnormally.

Recommended lubricants and capacities

To achieve proper engine and powertrain performance and durability, use only correct lubricants.
Correct lubricants promote engine efficiency that results in improved fuel economy.
These lubricants and fluids are for use in your vehicle.

Lubricant			Volume	Classification
Engine oil ¹ (drain and refill)	(Gasoline) 2.0 MPI		4.2 US qt. (4.0 L)	Full synthetic SAE 5W-20, API SM & ILSAC GF-4 (or above) or ACEA A5/B5
 TotalEnergies	(Gasoline) 1.6 T-GDi		4.2 US qt. (4.0 L)	Full synthetic SAE 5W-30, API Latest (ILSAC Latest) or ACEA A5/B5
	(Gasoline) 2.0 MPI	1.79 ~ 1.9 US qt. (1.7 ~ 1.8 L)		SAE 70W, API GL-4, HK SYN MTF 70W (SK), SPIRAX S6 GHME 70W MTF (H.K.SHELL), GS MTF HD 70W (GS CALTEX), Kia genuine MTF&DCTF70W SYNTHETIC
Manual transmission (MT) fluid ²	(Gasoline) 1.6 T-GDi			
Dual clutch transmission (DCT) fluid ²	(Gasoline) 1.6 T-GDi		2.01~2.11 US qt. (1.9~2.0 L)	HK D DCTF TGO-10 PLUS (SK), SPIRAX S6 GHDE 70W DCTF PLUS (H.K.SHELL), Kia genuine DCTF 70W SYNTHETIC PLUS
Intelligent Variable Transmission (IVT) fluid ²	(Gasoline) 2.0 MPI		7.1 US qt. (6.7 L)	MICHANG SP-CVT1, Kia Genuine ATF SP-CVT1 ²
Coolant ³	(Gasoline) 2.0 MPI	M/T	5.9 US qt. 5.6 L	A Phosphate based ethylene glycol based coolant
		IVT	5.9 US qt. (5.6 L)	
	(Gasoline) 1.6 T-GDi	M/T	7.3 US qt.	
		DCT	(6.9 L)	
Brake/clutch fluid ⁴			0.7 ~ 0.8 US qt. (0.7~0.8 L)	SAE J1704 DOT-4 LV, ISO4925 CLASS-6,FMVSS116 DOT-4
Fuel			14.0 US gal. (53 L)	-

*1. Refer to "Recommended SAE viscosity(thickness) number" on page 9-8.
*2. If the genuine oil that is developed for best performance is not used, it may cause the problems of transmission performance.
*3. Different type of coolant or water may damage the electrical component.
*4. To maintain best brake and ABS/ESC performance, use Kia genuine brake fluid.

Recommended SAE viscosity(thickness) number

⚠ CAUTION

Always clean the area around any filler plug, drain plug, or dipstick before checking or draining any lubricant. This is especially important in dusty or sandy areas and when the vehicle is used on unpaved roads. Cleaning the plug and dipstick areas will prevent dirt and grit from entering the engine and other mechanisms that could be damaged.

Engine oil viscosity has an effect on fuel economy and cold weather engine start and oil flowability. Lower viscosity engine oils can provide better fuel economy and cold weather performance. Higher viscosity engine oils are required for satisfactory lubrication in hot weather.

Using oils of any viscosity other than those recommended could result in engine damage.

When choosing an oil, consider the range of temperature your vehicle will be operated in before the next oil change. Select the recommended oil viscosity from the chart.

* NOTICE

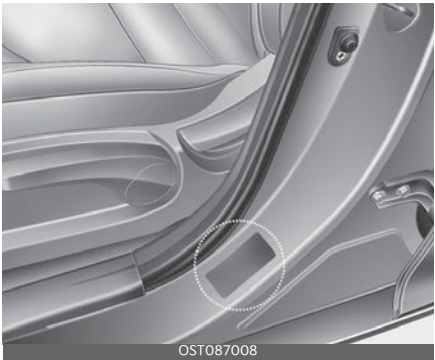
Never add any additives to the engine oil. Engine oil additives can change its the properties and may cause engine failure.

Temperature Range for SAE Viscosity Numbers										
Temperature	°C	-30	-20	-10	0	10	20	30	40	50
	°F	-10	0	20	40	60	80	100	120	
(Gasoline) 1.6 T-GDi						20W-50				
						15W-40				
					10W-30					
				5W-30, 5W-40						
(Gasoline) 2.0 MPI					10W-30					
				5W-20, 5W-30						



An engine oil displaying this American Petroleum Institute (API) Certification Mark conforms to the International Lubricant Specification Advisory Committee (ILSAC). It is recommended to only use engine oils that uphold this API Certification Mark.

Tire specification and pressure label



The tires supplied on your new vehicle are chosen to provide the best performance for normal driving.

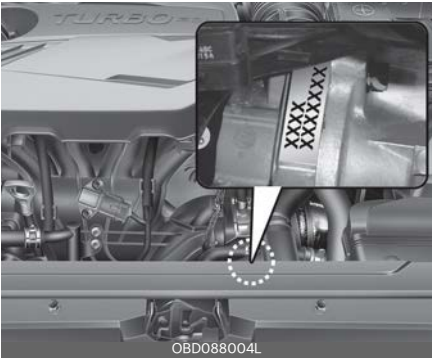
The tire label located on the driver's side center pillar gives the tire pressures recommended for your vehicle.

Engine number

(Gasoline) 2.0 MPI



(Gasoline) 1.6 T-GDi



The engine number is stamped on the engine block as shown in the drawing.